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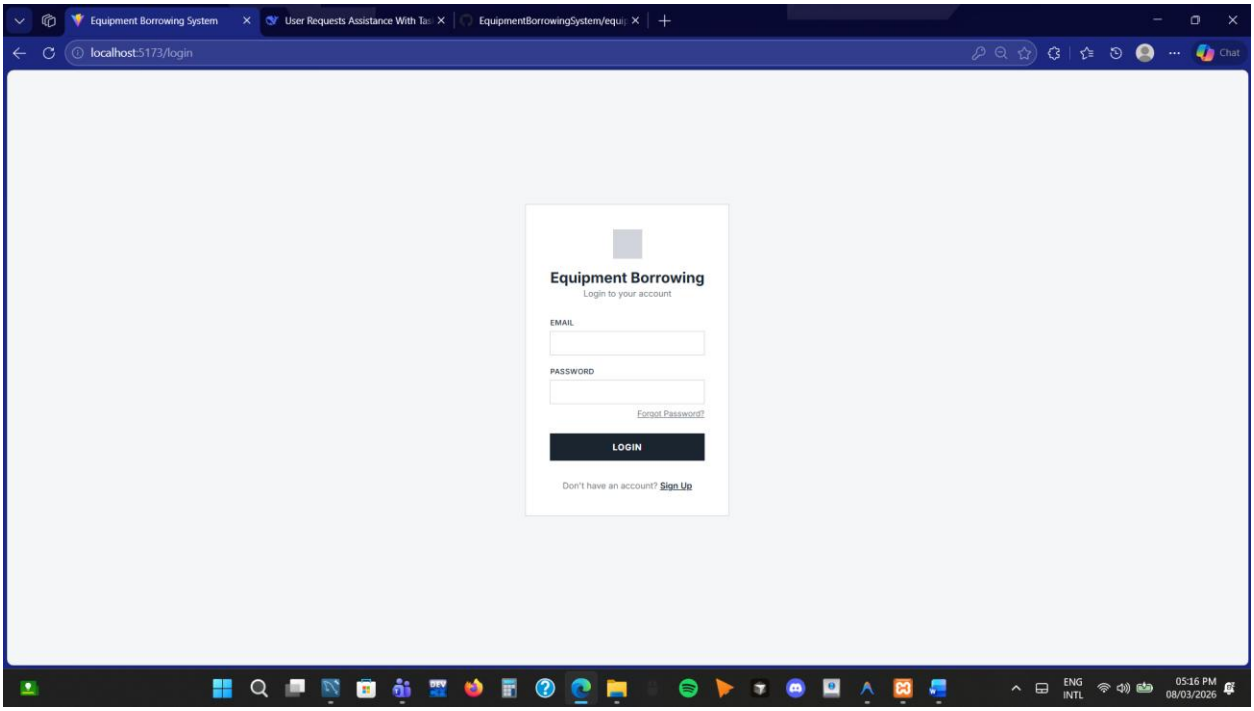

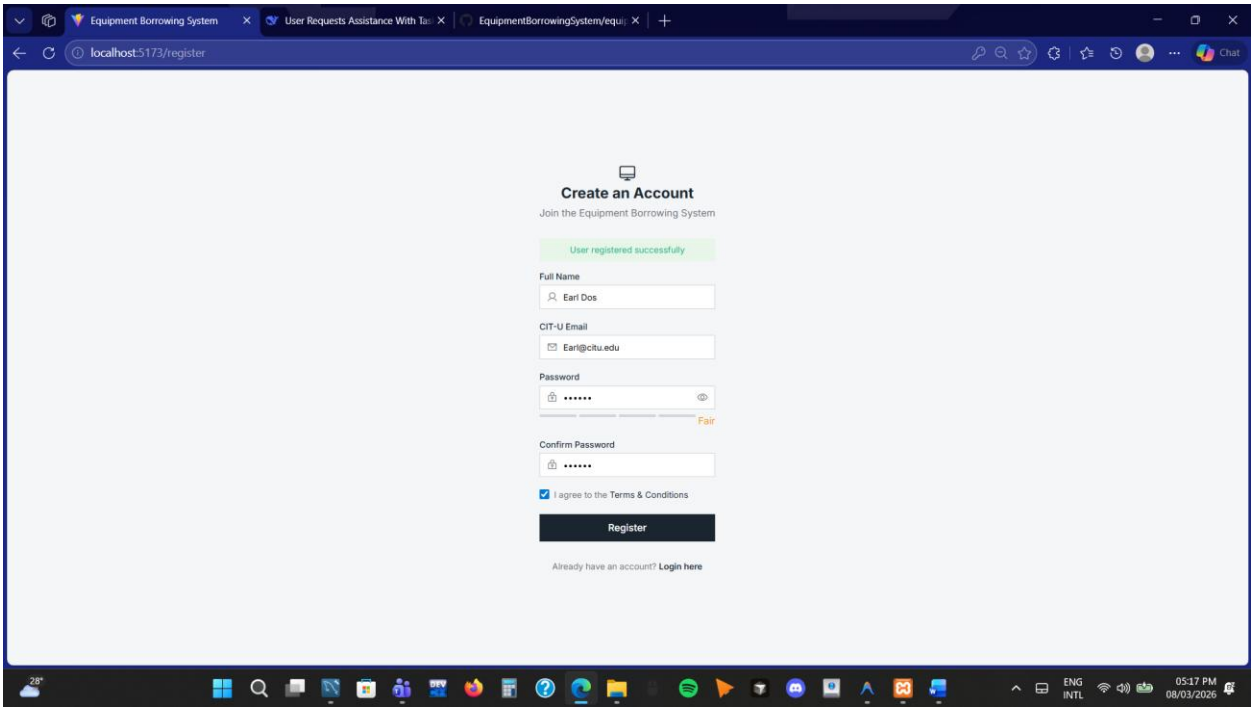

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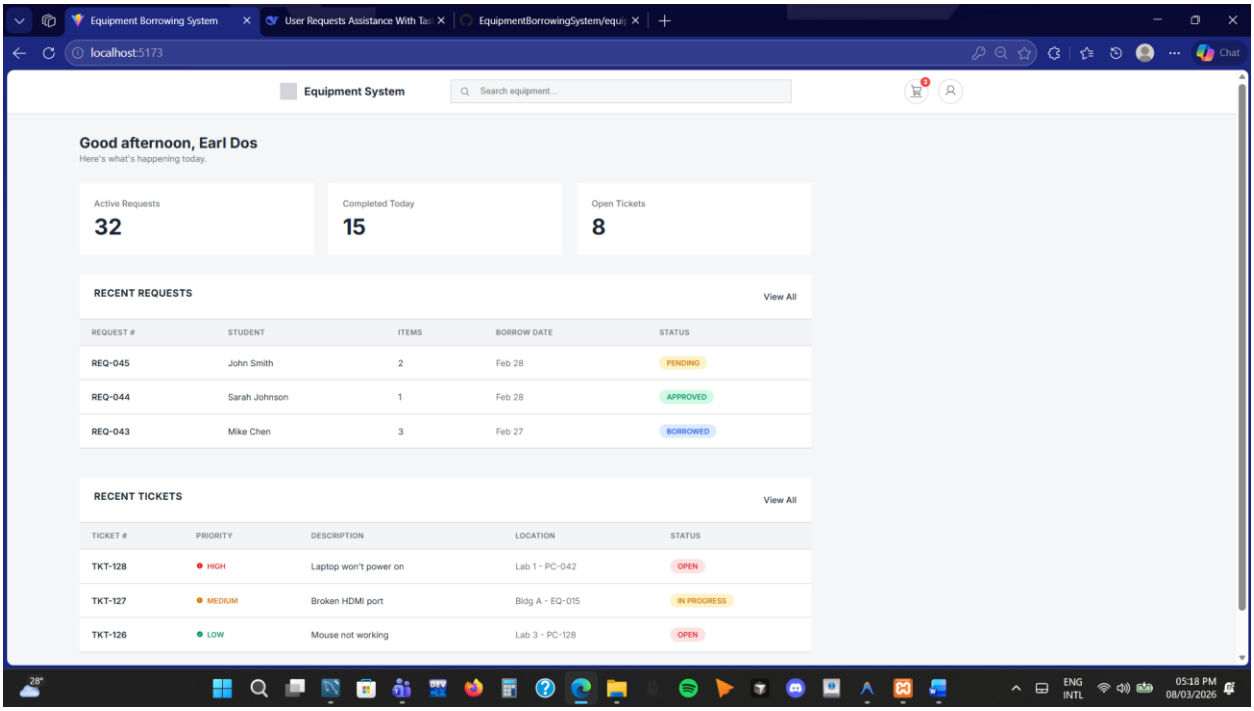

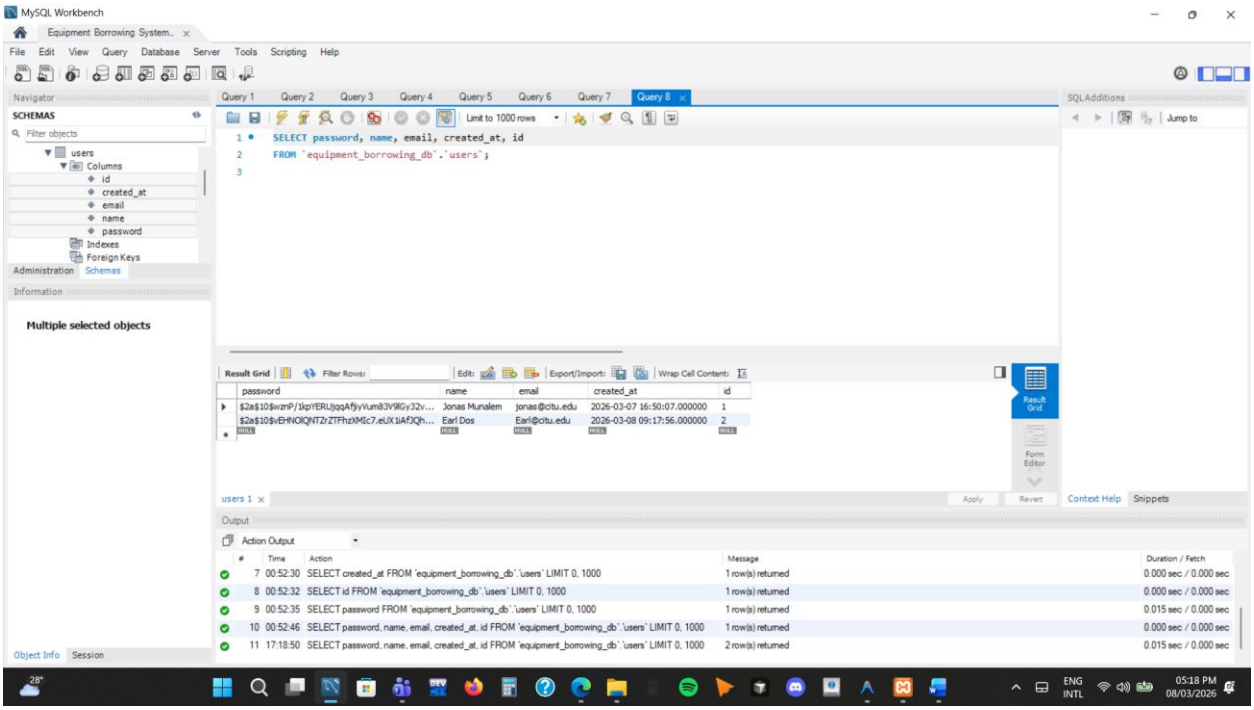

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EQUIPMENT BORROWING SYSTEM (EBS)

PHASE 1: IMPLEMENTATION SUMMARY

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1. REGISTRATION

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* Fields Used:

  The frontend registration form captures the user's Full Name,

  CIT-U Email, Password, and Confirm Password.

* Validation Process:

  - Frontend: React ensures all required fields are filled,

    passwords match, and tracks password strength.

  - Backend: Spring Boot utilizes Java Validation (@NotBlank,

    @Email, @Size) to enforce data integrity.

* Duplicate Prevention:

  A findByEmail() JPA Query checks the database upon submission.

  If the email exists, the backend returns an HTTP 400 Error,

  preventing duplicate accounts.

* Password Storage:

  Plain-text passwords are never saved. Spring Security

  encrypts the password using the BCryptPasswordEncoder

  hashing algorithm before saving it to the database.

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2. LOGIN

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* Credentials Used:

  The user provides their CIT-U Email and Password.

* Verification Process:

  The backend retrieves the User entity matching the email.

  The BCryptPasswordEncoder.matches() function compares the

  raw input against the stored hash to grant access.

* Post-Login Behavior:

  Upon success, the backend sends back user data

React stores this state and routes the user

  to the Dashboard based on their assigned role.

3. DATABASE TABLE

Table: users

Column	Data Type	Constraints	Description
id	BIGINT	Primary Key, Auto-Inc	Unique identifier for each user.
name	VARCHAR(255)	Not Null	User's full legal name.
email	VARCHAR(255)	Unique, Not Null	CIT-U email address for login.
password	VARCHAR(255)	Not Null	BCrypt encrypted password hash.
role	VARCHAR(50)	Not Null	User permissions (ADMIN/USER).
created_at	TIMESTAMP	Default Current	Account creation timestamp.

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4. API ENDPOINTS

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* POST /api/auth/register

  - Payload: { name, email, password }

  - Behavior: Validates, hashes, and creates record.

* POST /api/auth/login

  - Payload: { email, password }

  - Behavior: Verifies BCrypt hash and authorizes session.

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